PROJECT REPORT FOR INTERNSHIP

On

“Enterprise Customer Churn Prediction & Advanced Artificial Intelligence Analytics”

GOVERNMENT COLLEGE, DHOLPUR

Submitted in Partial Fulfillment for the Award of the Degree of

Bachelor of Science (B.Sc.)

By

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Project Summary

This project demonstrates a reproducible customer churn analytics workflow using Python. It includes data quality checks, descriptive statistics, visualisation, preprocessing, model training, and evaluation of a Logistic Regression classifier.

Prepared by: JATIN | Government College, Dholpur | AICTE / IBM SkillsBuild Data Analytics with AI Internship 2026.

1. Introduction

Customer churn occurs when a customer stops using a service. Analysing churn-related information can help an organisation identify patterns worth investigating and plan customer-retention research. Predictive models can estimate which records resemble historical churn patterns, but predictions should be assessed carefully before operational use.

2. Problem Statement

The project demonstrates how customer attributes such as tenure, monthly charges, contract type, internet service, and technical-support status can be organised and used to build a basic churn-classification workflow.

3. Objectives

- Inspect data structure, missing values, and duplicate rows.
- Summarise churn labels, monthly charges, and customer tenure.
- Compare churn distribution across contract categories.
- Prepare numerical and categorical variables for modelling.
- Train a Logistic Regression classification pipeline.
- Evaluate predictions with accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix.
- Document limitations and responsible use of the results.

4. Tools and Technologies

Python, Pandas, NumPy, Matplotlib, and scikit-learn. The script can run locally or in a compatible Google Colab environment.

5. Dataset and Limitation

The supplied script generates 1,200 synthetic practice records with a fixed random seed. The data are simulated for educational demonstration; they are not actual customer records and are not taken from a telecom company. The churn patterns are partly built into the data generator, so model scores only demonstrate that the pipeline runs and must not be interpreted as real-world performance.

6. Methodology

1. Generate a reproducible synthetic dataset with tenure, monthly charges, contract type, internet service, technical-support status, and churn label.
2. Save the practice dataset and check dimensions, missing values, and duplicate records.
3. Calculate descriptive statistics including churn rate, average monthly charges, and average tenure.
4. Separate input features from the churn target and use a stratified train/test split.

5 Impute/scale numerical variables and impute/one-hot encode categorical variables.

6 Train a class-balanced Logistic Regression model in a scikit-learn pipeline.

7 Evaluate with accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix.

8 Save summary tables and charts in an outputs folder.

7. Expected Outputs

When the script runs successfully, it prints data-quality checks and descriptive metrics, then creates an outputs folder containing the generated synthetic dataset, evaluation_metrics.json, churn_rate_by_contract.csv, confusion_matrix.png, and churn_by_contract.png. Exact metrics should be copied into final findings only after running the script.

8. Evaluation Measures

Accuracy measures the share of correct predictions. Precision measures how many records predicted as churn are labelled churn in the test data. Recall measures how many churn-labelled records are detected. F1-score balances precision and recall. ROC-AUC evaluates how well predicted scores rank the two classes across thresholds. These measures are meaningful for real use only when evaluated on an appropriate, representative, approved dataset.

9. Responsible Use and Limitations

- The included data are synthetic and do not support claims about a real company.
- A real submission may require a dataset supplied or approved by the internship mentor.
- For real data, cite the dataset source and document permission, cleaning decisions, missing values, class imbalance, and exclusions.
- Avoid data leakage and evaluate on a held-out test set.
- Do not use a demo model to make actual customer decisions.

10. Conclusion

The project provides a clear example of a customer churn analytics and machine-learning pipeline, from data inspection through model evaluation. It demonstrates reproducible Python practice and basic classification metrics. Real-world conclusions require an approved and documented dataset, careful validation, and an updated report based on the resulting evidence.

11. Future Scope

- Replace synthetic data with an approved, documented dataset.
- Compare Logistic Regression with suitable baseline models.
- Evaluate precision-recall trade-offs and calibration.
- Add feature-importance or model-explanation methods.
- Build a simple dashboard if required by the internship.

Appendix: Running the Project

Install dependencies with pip install -r requirements.txt, then run python churn_prediction.py. The script creates its output files in an outputs directory. See README.md for full instructions.